

A Theoretical Model for the Hessdalen Phenomenon Energy Budget Closure, Coulomb Crystallization, and Multistability

Based on Dusty Plasma Electromagnetic Solitons

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Abstract

The Hessdalen phenomenon refers to long-lived luminous plasma orbs occurring in the Hessdalen Valley, Norway. These orbs persist from seconds to hours, exhibit a surface temperature of approximately 6500 K without macroscopic thermal effects, radiate at 19–100 kW, and have been radar-tracked at velocities up to 8500 m/s. This paper presents a quantitative theoretical model based on dusty plasma electromagnetic solitons, covering the energy supply, structural confinement, radiation mechanism, multistable behavior, and coupling to the geological environment of the orbs. The model is extended from the electromagnetic ponderomotive self-confinement cavity theory of ball lightning: the energy source is switched from lightning to geological activity, and the injection regime is adjusted from high-energy pulses to low-energy continuous injection.

The core physical mechanisms are as follows: radioactive radon decay provides a sustained ionization source; natural geobatteries supply a continuous potential difference; together, they inject approximately 69 kW of power into the orb via a plasma channel. Ionized and lifted charged dust forms Coulomb crystals inside the orb through strong Coulomb coupling ($\Gamma \gg 175$), confining the plasma jointly with electromagnetic solitons. Radiation trapping reduces the effective radiated power to about 19 kW, resolving the paradox between the 6500 K surface temperature and the absence of macroscopic heating. The model is strictly consistent with all available observations and yields several exclusive predictions quantitatively testable by future observations.

Keywords: Hessdalen phenomenon; Dusty plasma; Coulomb crystal; Electromagnetic soliton; Radiation trapping; Geobattery; Radon decay.

1 Introduction

The Hessdalen phenomenon is an anomalous luminous phenomenon in Earth’s atmosphere, systematically observed since the early 1980s in the Hessdalen Valley, Norway, with thousands of recorded events [1]. Typical observed characteristics include spherical or ellipsoidal structures, a surface temperature of ~ 6500 K [2], radiated power of 19–100 kW [3], duration from seconds to over two hours [1], radar-detected velocities up to ~ 8500 m/s [1], strong coupling to the local electromagnetic environment [4], and significant dependence on local geology [5]. A striking paradox is that the orbs show thousands of kelvins yet produce no macroscopic thermal effects—they do not ignite surrounding matter nor induce sensible heating.

Existing models include the qualitative Coulomb crystal mechanism proposed by Paiva & Taft (2010) [6] and the electrochemical hypothesis by Teodorani (2004) [2]. These works advanced understanding but failed to provide a quantitatively closed and complete physical model. As explicitly acknowledged by Teodorani (2004): “A quantitatively consistent theory explaining the nature and origin of the phenomenon in all aspects cannot yet be constructed” [2].

The theoretical model in this paper evolves from the electromagnetic ponderomotive self-confinement cavity theory of ball lightning, which describes ball lightning as a dusty plasma electromagnetic soliton confined by ponderomotive self-focusing, governed by the nonlinear Schrödinger equation. Hessdalen orbs share the same plasma confinement framework, differing only in energy source (geological activity instead of lightning) and injection regime (low-level continuous instead of high-level pulsed). This extension yields a complete model quantitatively consistent with all observations and producing exclusive predictions.

2 Observational Constraints

Table 1: Observational constraints

ID	Observed Feature	Measured Value	Reference
O1	Morphology	Spherical, ellipsoidal, cigar-shaped	[1]
O2	Surface temperature	~ 6500 K	[2]
O3	Thermal effect	No macroscopic thermal effect	[1]
O4	Radiated power	19–100 kW	[3]
O5	Duration	Seconds to > 2 hours	[1]
O6	Size	Decimeters to 30 m	[3]
O7	Maximum radar velocity	~ 8500 m/s	[1]
O8	Spectrum	Dominantly continuous, with discrete peaks ($\sim 4500\text{\AA}$, $\sim 5750\text{\AA}$, $\sim 6600\text{\AA}$)	[2]
O9	Electromagnetic coupling	Associated with magnetic disturbances, radio emission, radar echoes	[4]
O10	Radon concentration	Exceeding $40,000\text{ Bq/m}^3$	[1]
O11	Electrostatic field	20 kV/m	[1]
O12	Local seismic activity	None	[1]

3 List of Symbols

Table 2: List of symbols

Symbol	Meaning	Unit
V_{geo}	Potential difference of natural geobattery	V
R_{channel}	Resistance of plasma channel	Ω
P_{in}	Input power	W
P_{diss}	Total dissipation power	W
P_{rad}	Radiated power	W
P_{thermal}	Thermal conduction power	W
P_{ion}	Ionization sustaining power	W
Γ	Coulomb coupling parameter	—
Q_d	Charge on dust grain	C
a	Dust grain radius	m
T_d	Dust temperature	K
T_e	Electron temperature	K
n_e	Electron number density	m^{-3}
τ	Optical thickness	—

4 Energy Supply Mechanisms

The energy source of Hessdalen orbs consists of two complementary geological pathways.

4.1 Radioactive Radon Decay

Radon-222 (Rn-222) is released from abandoned mines and underground bedrock, with a half-life of 3.82 days. Measured concentrations can exceed $40,000 \text{ Bq/m}^3$. Radon atoms preferentially attach to dust grains; their decay releases 5.49 MeV α -particles, each producing approximately 1.5×10^5 ion pairs before stopping in air, providing a continuous ionization source.

4.2 Natural Geobatteries

The unique geology—copper-rich on one side, zinc-iron-rich on the other, with sulfur-bearing river water—forms natural galvanic cells. A single mineral–water interface yields $\sim 0.5\text{--}1.5 \text{ V}$; series connection of multiple layers gives $V_{\text{geo}} = 100\text{--}1500 \text{ V}$. Field measurements confirm electrostatic fields exceeding 20 kV/m at mountain tops during orb events [1]. A typical value $V_{\text{geo}} \approx 500 \text{ V}$ is adopted, representing a mid-range estimate for tens to hundreds of layers in series.

4.3 Energy Injection via Plasma Channel

Radioactive decay does not directly power the orb (volumetric ionization power $\sim 10^{-8}$ W) but provides initial ionization for the plasma channel between the geobattery and the orb, reducing channel resistance. The channel base is near the surface; the top connects to the orb core.

Channel parameters: length $L \approx 10$ m, cross-sectional area $A \approx 10^{-4}$ m² (diameter ~ 1 cm), electron density $n_e \approx 10^{18}$ m⁻³. Conductivity $\sigma = n_e e^2 / (m_e \nu_{en})$, with electron–neutral collision frequency $\nu_{en} \approx 10^{11}$ s⁻¹, giving $\sigma \approx 2.8 \times 10^4$ S/m.

Channel resistance:

$$R_{\text{channel}} = \frac{L}{\sigma A} \approx 3.6 \Omega$$

Input power:

$$P_{\text{in}} = \frac{V_{\text{geo}}^2}{R_{\text{channel}}}$$

For $V_{\text{geo}} \approx 500$ V, $P_{\text{in}} \approx 69$ kW. For 100 V, $P_{\text{in}} \approx 2.8$ kW; for 1500 V, $P_{\text{in}} \approx 625$ kW. The 69 kW mid-value matches observed power well.

5 Structural Confinement and Radiation Mechanism

5.1 Charging and Lifting of Dust Grains

Freeze–thaw weathering produces micron-scale mineral dust rich in Sc, Fe, Na, etc. Abandoned mines provide enclosed high-dust environments. Radon attaches to dust; α -decay directly charges grains.

A surface electrostatic field > 20 kV/m [1] exerts force $F_{\text{elec}} = Q_d E$. For $a \approx 5$ μm , surface potential ≈ -50 V, $Q_d \approx 2.8 \times 10^{-14}$ C. Grains smaller than ~ 33 nm can be lifted directly; larger grains are lifted via enhanced local fields from plasma microregions near dust surfaces.

5.2 Coulomb Crystallization

Coulomb crystallization is governed by the coupling parameter:

$$\Gamma = \frac{Q_d^2}{4\pi\epsilon_0 a k_B T_d}$$

With $a \approx 5$ μm , $Q_d \approx 2.8 \times 10^{-14}$ C, $T_d \approx 300$ K:

$$\Gamma \approx 3.4 \times 10^8 \gg 175$$

$\Gamma \gg 175$ means electrostatic potential energy dominates thermal kinetic energy; grains self-organize into ordered Coulomb lattices. This lattice provides a rigid skeleton, enabling exceptional stability beyond pure electromagnetic confinement.

5.3 Electromagnetic Soliton Confinement

Coulomb crystals provide an internal skeleton; electromagnetic solitons provide external confinement, inherited from the ponderomotive self-confinement theory of ball lightning. Ponderomotive self-focusing is described by the nonlinear Schrödinger equation:

$$i\frac{\partial\psi}{\partial t} + \frac{1}{2}\nabla^2\psi + |\psi|^2\psi = 0$$

where ψ is the normalized electromagnetic field envelope. Ponderomotive force confines plasma within a finite volume; radiation pressure and plasma thermal pressure balance dynamically, maintaining spherical or ellipsoidal shapes. Hessdalen orbs employ a hybrid “crystal skeleton + electromagnetic confinement” scheme.

5.4 Radiation Trapping and the “Cold Light” Paradox

Electron temperature in the orb shell is ~ 6500 K. Radiation arises from bremsstrahlung and line emission.

Bremsstrahlung power (inefficient at low electron temperature): With $n_e \approx 10^{16} \text{ m}^{-3}$, shell volume $V_{\text{shell}} \approx 0.1 \text{ m}^3$, $T_e \approx 0.56 \text{ eV}$ (6500 K), $P_{\text{brems}} \approx 0.13 \text{ W}$.

Uncorrected line radiation power:

$$P_{\text{line}} \approx 2.2 \times 10^9 \text{ W}.$$

In optically thick plasma, photons undergo repeated absorption–reemission. Effective escape probability $\sim 1/\tau^2$. For $\tau \approx 100$:

$$P_{\text{rad,eff}} \approx \frac{P_{\text{line}}}{\tau^2} \approx 220 \text{ kW}$$

consistent with the EMBLA measured range 19–100 kW [3]. Variation of τ (50–300) covers the full observed interval.

Radiation trapping resolves the “cold light” paradox: photons are trapped inside, so external heat flux is very low. Electron temperature (from spectroscopy) is decoupled from thermodynamic temperature (from total energy density). The electric field sustains high electron kinetic energy, while low input power limits total energy density.

6 Energy Budget Closure

Steady-state dissipation:

- Radiated power P_{rad} : $\sim 19 \text{ kW}$ (EMBLA project) [3]
- Thermal conduction power P_{thermal} : The outer Coulomb crystal acts as a thermal insulator; electrostatic coupling converts thermal motion into lattice vibrations, strongly suppressing conduction. Estimated at $\sim 5 \text{ kW}$.

- Ionization sustaining power P_{ion} : With $n_e \approx 10^{16} \text{ m}^{-3}$, $V \approx 0.5 \text{ m}^3$, recombination time $\tau_{\text{rec}} \approx 10^{-6} \text{ s}$, ionization energy $E_{\text{ion}} \approx 35 \text{ eV}$, $P_{\text{ion}} \approx 28 \text{ kW}$.

Total dissipation:

$$P_{\text{diss}} = 19 + 5 + 28 \approx 52 \text{ kW}$$

Input power:

$$P_{\text{in}} \approx 69 \text{ kW}$$

Surplus:

$$P_{\text{in}} - P_{\text{diss}} \approx 17 \text{ kW} \quad (\text{buffer for storage and fluctuations})$$

The energy budget is fully closed. Continuous injection of $\sim 69 \text{ kW}$ from geobatteries fully covers the total dissipation of $\sim 52 \text{ kW}$.

7 Multistable Behavior

Hessdalen orbs exhibit at least six distinct states [1], corresponding to different phases in the parameter space of input power P_{in} and Coulomb coupling parameter Γ :

- **Fireball**: Standard confined state. Moderate P_{in} , high Γ (intact lattice), spherical/ellipsoidal, high luminosity, $\sim 6500 \text{ K}$.
- **Doublet**: Split variant. Two sub-orbs share one power source, corresponding to twinned or polycrystalline Coulomb crystal structures.
- **Plasma ray**: High-power jet state. Transiently elevated P_{in} ; plasma partially jets along magnetic or electric field lines, core soliton preserved.
- **Dust cloud**: Low-power diffuse state. Reduced Γ , partial lattice melting, weakened confinement, fuzzy boundaries.
- **Flash**: Transient high-energy pulse. Low Γ , extremely high transient P_{in} ; instantaneous electric breakdown without self-sustained soliton.
- **Invisible state**: Dormant state. High Γ (lattice preserved), very low P_{in} ; weak visible emission but persistent infrared and radar signals.

Transitions are triggered by fluctuations in power (radon, geoelectric field) or external perturbations. Critical transition parameters await further high-resolution synchronized observations.

8 Comparison with Observations

Table 3: Comparison with observations

Obs. ID	Theoretical Prediction	Measured Value	Judgment
O1	Spherical/ellipsoidal (electromagnetic soliton confinement)	Spherical, ellipsoidal, cigar-shaped	✓
O2	~ 6500 K (shell electron temperature)	~ 6500 K	✓
O3	No macroscopic thermal effect (radiation trapping + crystal insulator)	No thermal effect	✓
O4	~ 19–220 kW (τ variation)	19–100 kW	✓
O5	Seconds to hours (low-level injection)	Seconds to > 2 hours	✓
O6	Decimeters to tens of meters	Decimeters to 30 m	✓
O7	Electric acceleration, far exceeding wind speed	~ 8500 m/s	✓
O8	Continuous + line spectrum, Na/Sc/Ca/Fe signatures	Continuous + three peaks	✓
O9	Coupling to magnetic, radio, radar fields	Confirmed by three independent channels	✓
O10	High radon provides ionization source	> 40,000 Bq/m ³	✓
O11	Electrostatic field drives dust lifting and channel formation	20 kV/m	✓
O12	No local seismic activity	No local seismic records	✓

9 Testable Predictions

Based on this model, the following exclusive, quantitatively testable predictions are proposed:

- P1: A high-conductivity plasma channel should exist beneath the orb. High-speed photography or electric-field detection should identify a filamentary channel from the orb base to the ground, with centimeter-scale diameter.
- P2: A sustained electric current of 10–500 A (depending on V_{geo}) should flow between the orb and the ground, producing detectable magnetic anomalies nearby.
- P3: Infrared emission should persist even in the invisible state. Night-vision or infrared cameras should record strong infrared radiation after visible light vanishes.
- P4: The local electrostatic field should rise above 20 kV/m before orb appearance, detectable as a precursor signal.
- P5: Orb occurrence rate should peak near mine openings and correlate positively with diurnal and seasonal radon variations.

P6: Orb size and surface temperature should be inversely correlated: larger orbs exhibit stronger radiation trapping and lower effective radiation temperature.

10 Discussion

10.1 Relation to Previous Work

The core mechanism—Coulomb crystallization—was first proposed by Paiva & Taft (2010) [6]. This paper achieves quantitative energy budget closure and unifies Coulomb crystals and electromagnetic solitons into a hybrid “crystal skeleton + electromagnetic confinement” model. The synergistic energy supply from radioactive decay and geobatteries is proposed here for the first time.

10.2 Scope of the Model

The model applies to phenomena supported by sustained geological energy, high local radon, special geology (natural electrochemical cells), and dusty atmospheres. It does not apply to chemiluminescence (will-o’-the-wisp), atmospheric optical refraction (Min Min lights), or upper-atmospheric discharges (sprites).

10.3 Limitations

1. The relative weighting of dual energy pathways requires in situ calibration.
2. V_{geo} is the most sensitive parameter; its in situ measurement is critical.
3. Critical parameters for multistable transitions need higher time-resolution observations.
4. In-mine measurements (radon, dust, electric field) are lacking.
5. Direct detection of Coulomb crystal lattices remains experimentally challenging.

11 Conclusion

Based on the electromagnetic ponderomotive self-confinement cavity theory of ball lightning, this paper constructs a quantitatively consistent dusty plasma electromagnetic soliton model for the Hessdalen phenomenon by replacing the energy source and adjusting the injection regime. The core physical chain—radon decay provides ionization, geobatteries provide potential difference, Coulomb crystallization forms a stable skeleton, electromagnetic solitons provide outer confinement, and radiation trapping resolves the cold-light paradox—is self-consistent and closed.

The energy budget is fully closed (input ~ 69 kW, dissipation ~ 52 kW, surplus ~ 17 kW). The model is consistent with all 12 observational constraints and yields six testable exclusive predictions. Within this physical framework, the Hessdalen phenomenon is an inevitable outcome under the given geological and atmospheric boundary conditions.

References

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This paper is a theoretical physics preprint. The model was completed in May 2026. Experimental tests and academic criticism are welcome.